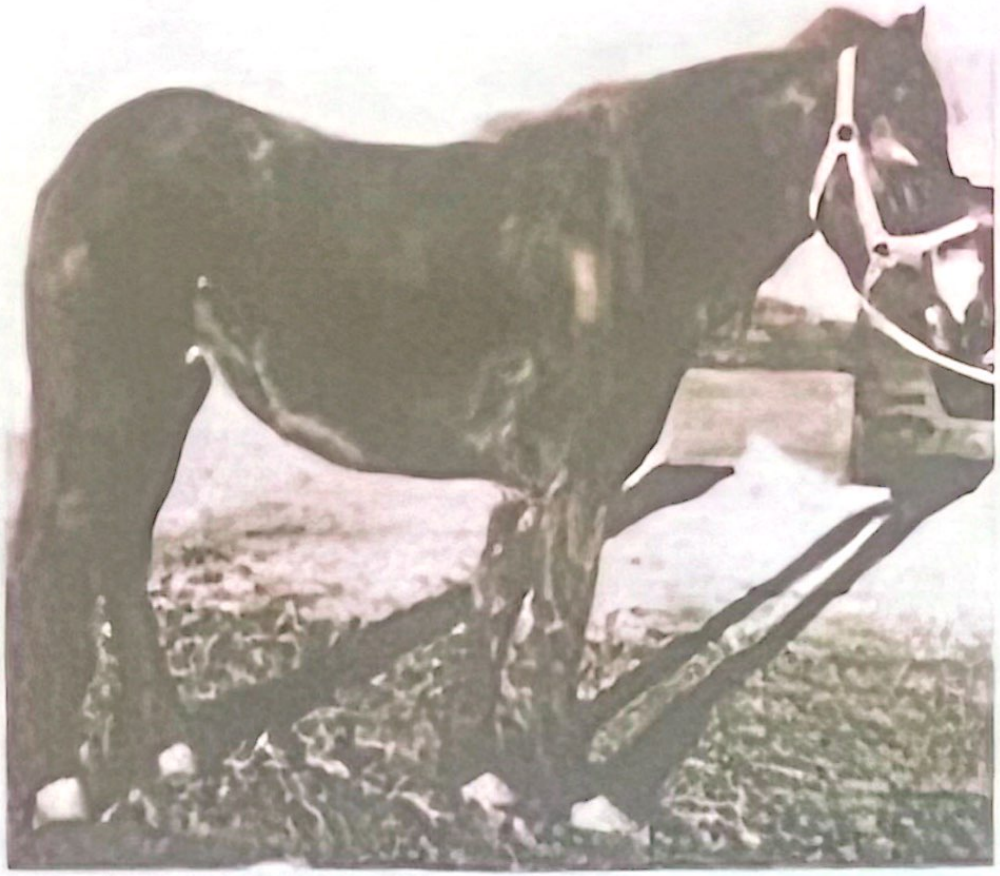
I- Monosomy (monosomic case)

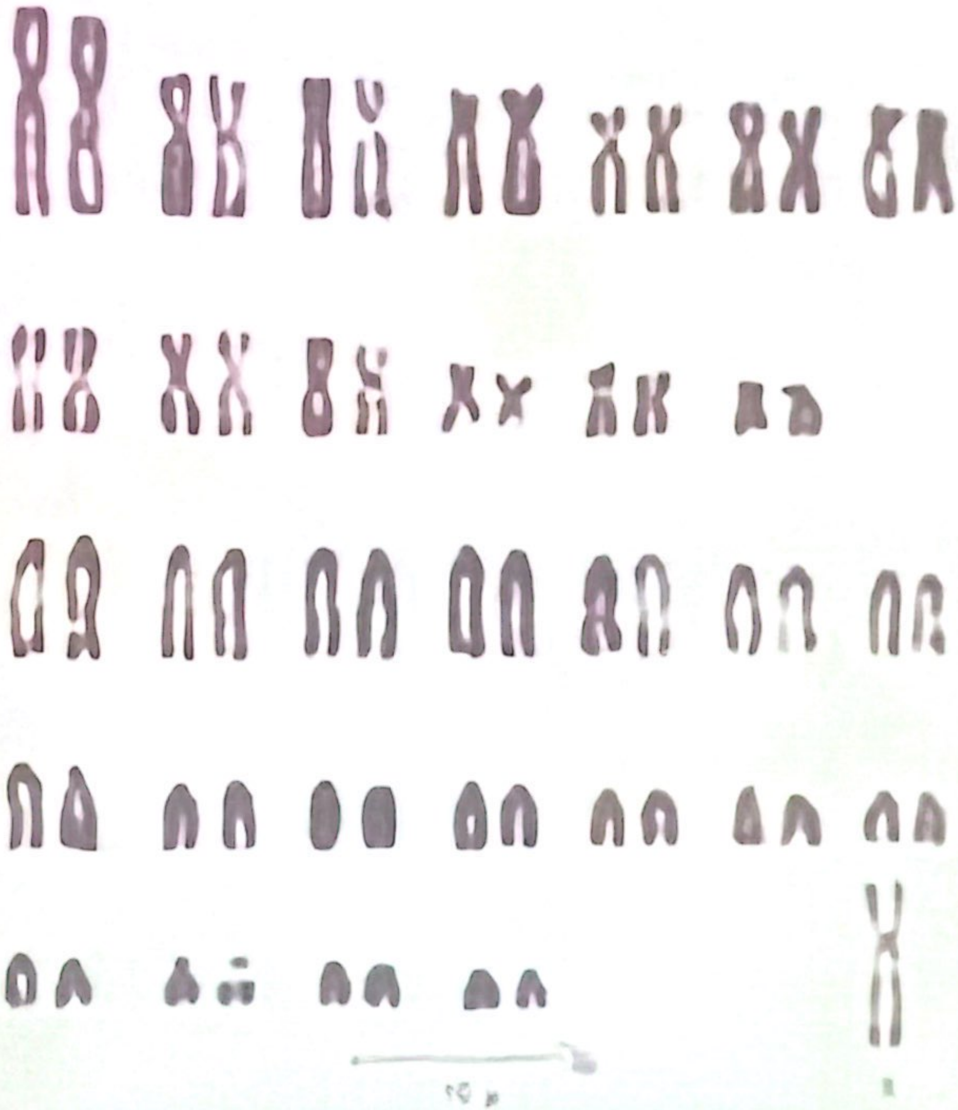
1. Turner Syndrome (63,XO)

1. Reported in mares
2. External appearance was generally normal in all of these mares.
3. Estrous cycles were either irregular or absent,
4. Uterus and ovaries are smaller than average,
5. Ovaries (smooth) often lacked follicles.

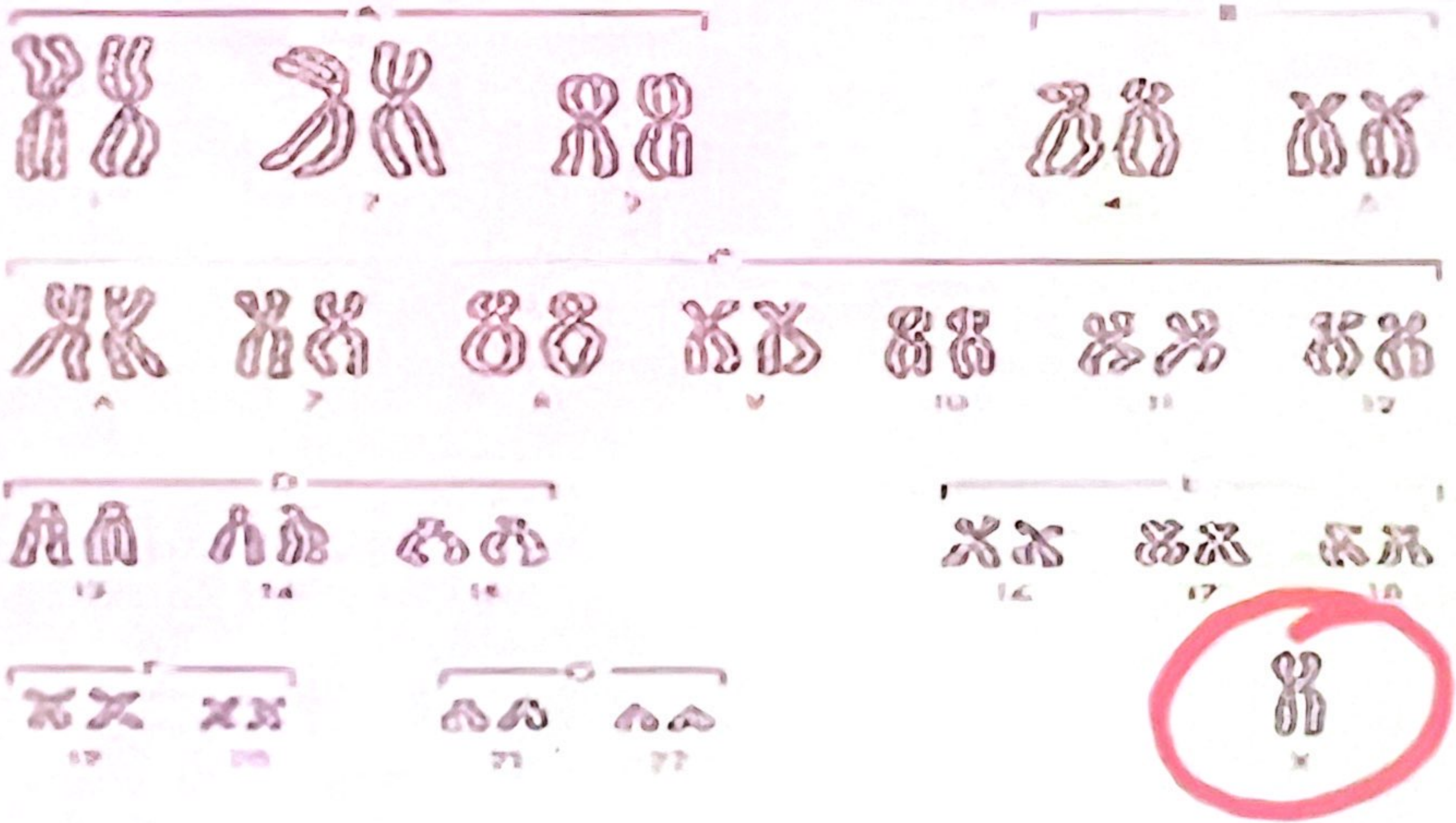
- There was a lack of one X chromosome.
- Their karyotype is written as 63, XO with the O including absence of an X chromosome
- XO individuals reported in mice, rats, cats, pigs and in humans "45, Xo"



XO-horse



Horse - karyotype, 2n = 63,XO



45, xo

II- Trisomy (trisomic case)

1. Klinefelter Condition in Cats (39, XXY)

Sterile males because of failure to produce sperms by the underdeveloped testes has unusual tortoise-shell colored tom cats.

caused by a pair of codominant X-linked genes,
one "O" gives orange color
"o" gives black colour.

The heterozygote "Oo" has a mixture of both colours, which, is called tortoise-shell,

Female		male	
xOxO	Orange	xOy	Orange
xoxo	black	xoy	black
xOxo	tortois shill color		

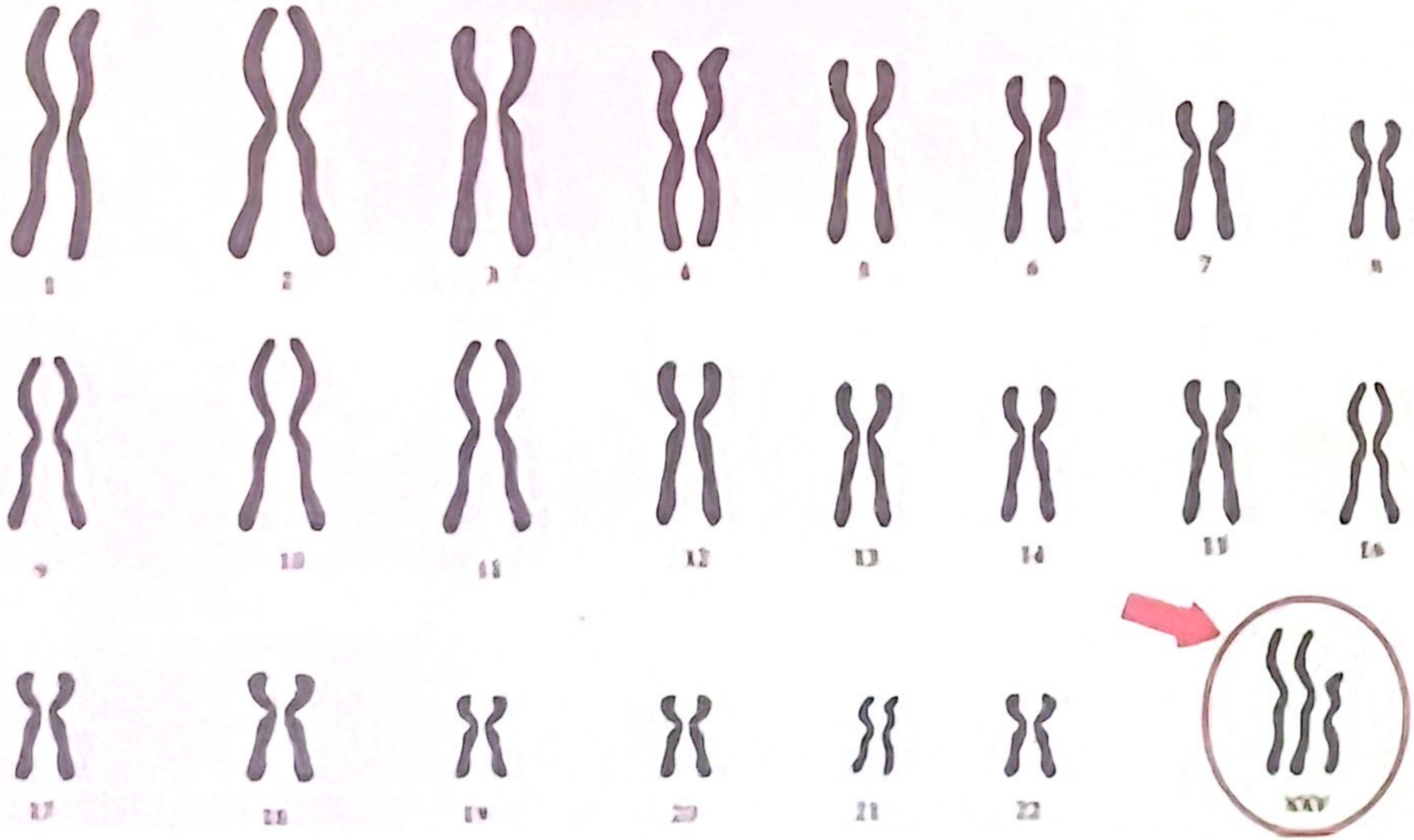
Tortoise-shell males was abnormal,
it was found to be a trisomy of X chromosome and is a typical case of Klinefelter condition,

39, XXY (instead of the normal 38,XY).

This would allow the heterozygous colour of tortoiseshell whose genotype would be OoY

47, XXY in Man

Klinefelter Syndrome



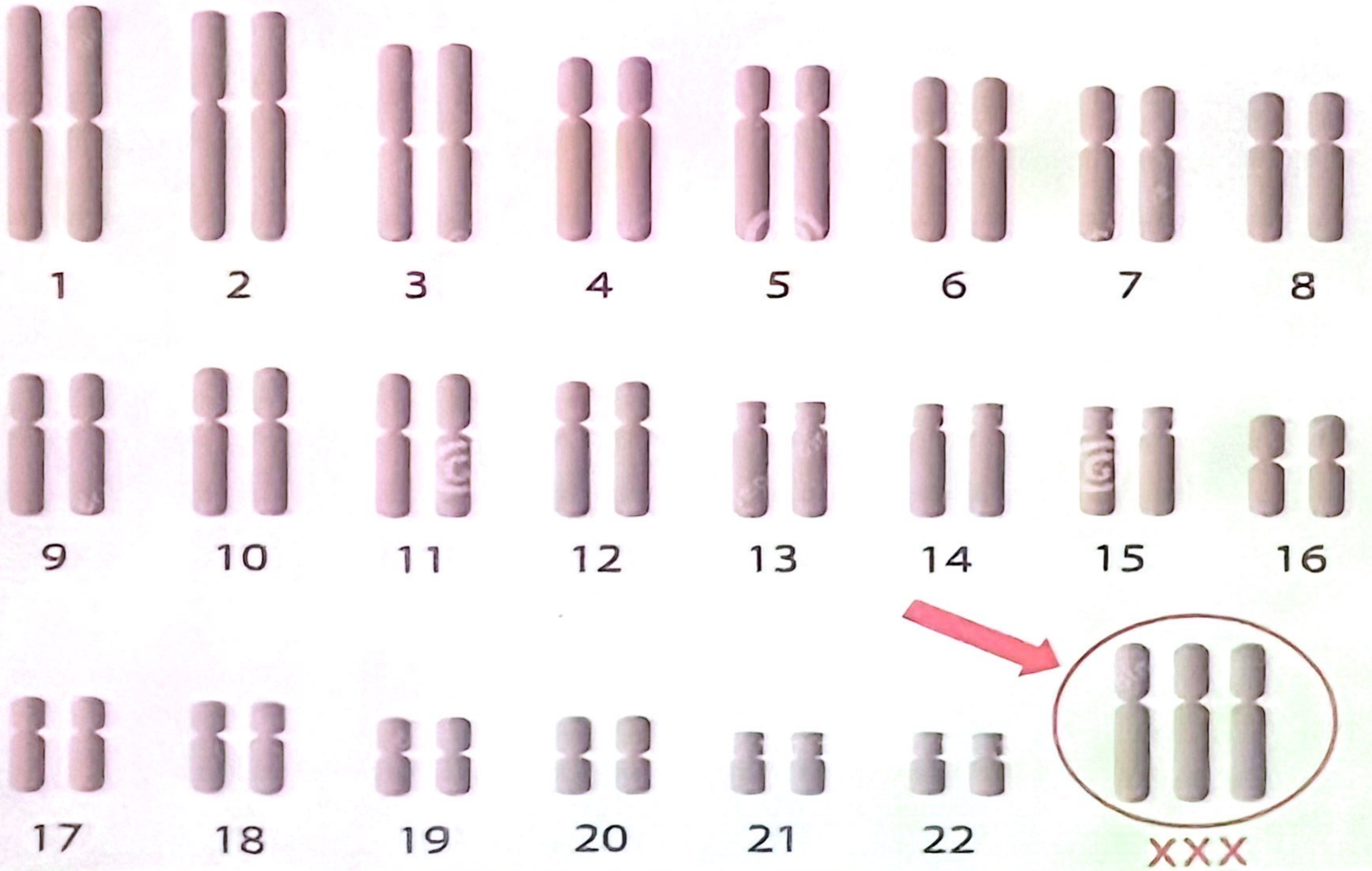
2. Triple-X Condition (65,XXX):

Reported in Mars with an extra X the condition is written as (65, XXX).

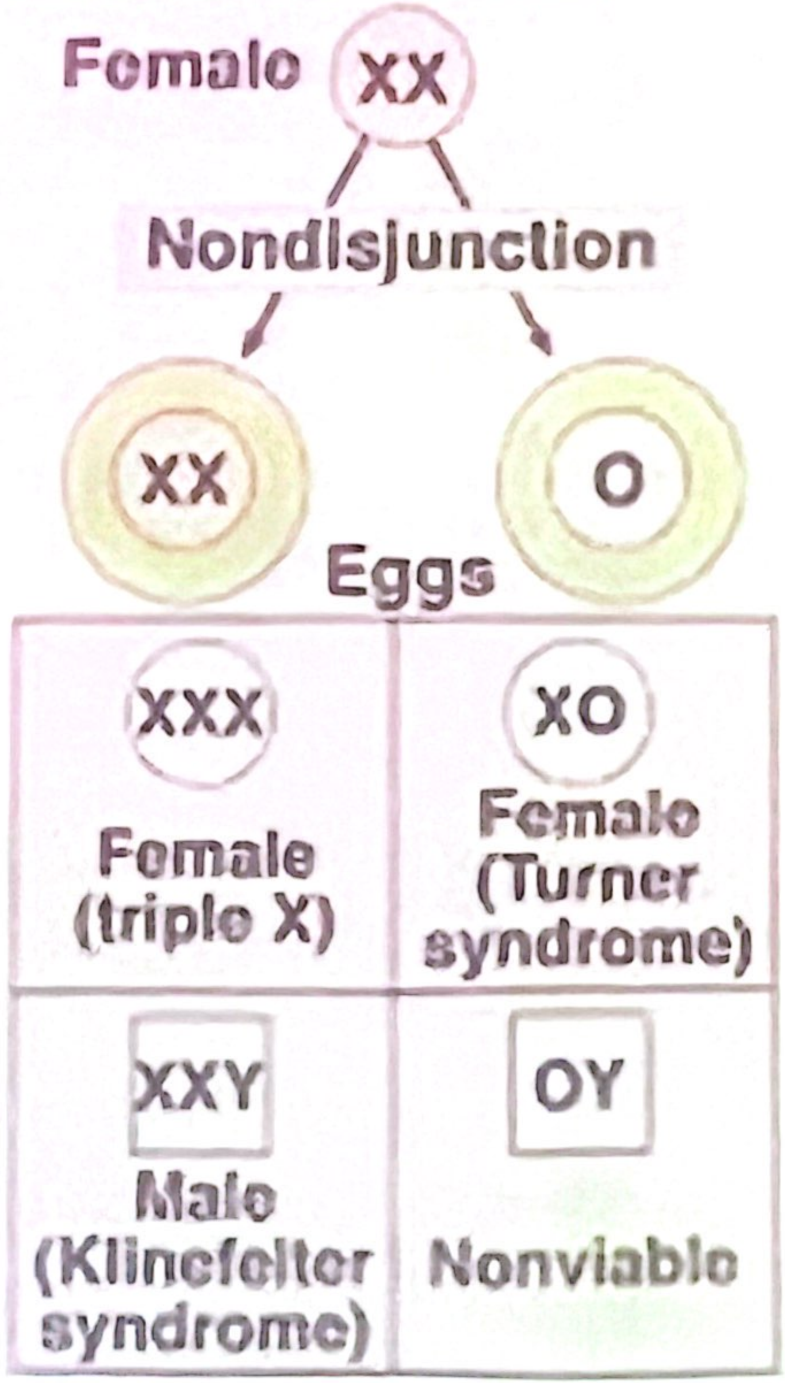
When examined cytologically, the cells contain
two Barr bodies
indicating that only one X chromosome in each cell remains active.

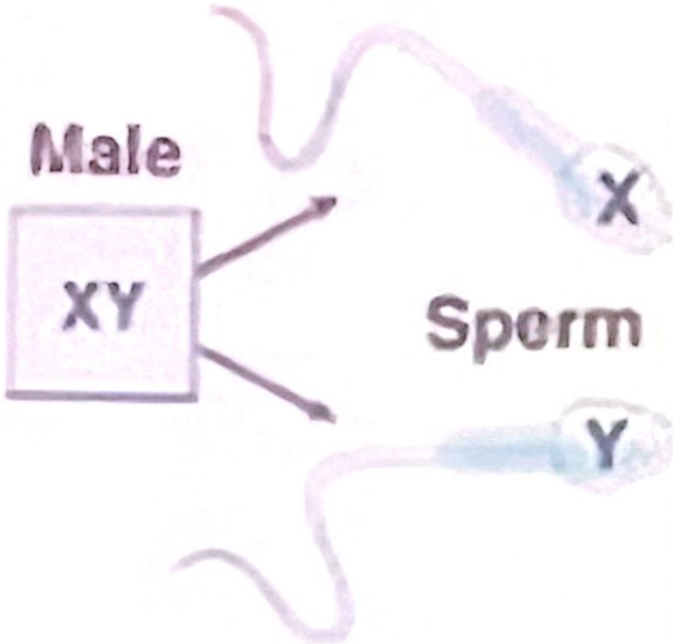
XXX individuals are mainly infertile,

Triple X Syndrome



Nondisjunction and Sex Chromosomes





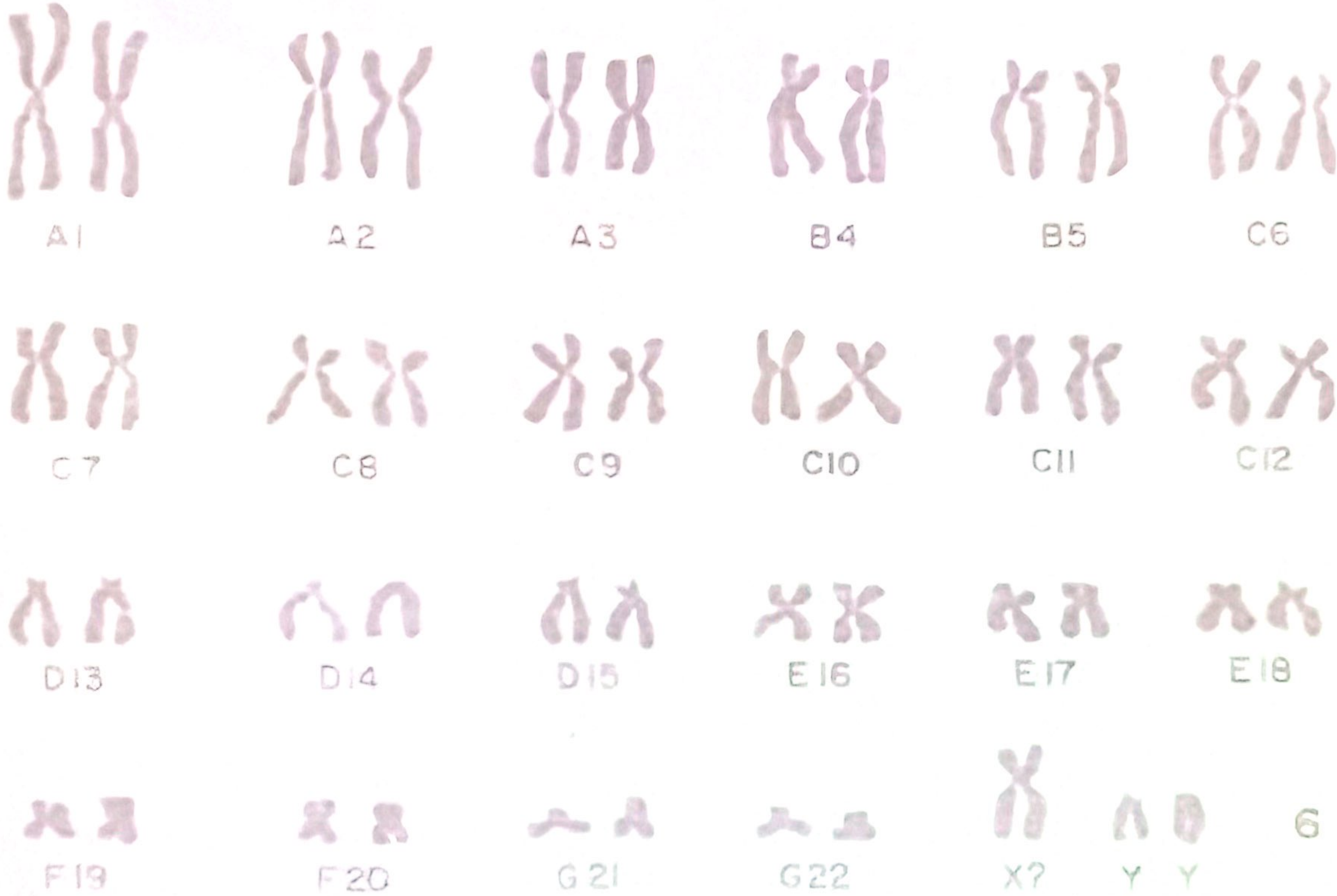
3. The XYY Male Condition (47,XYY):

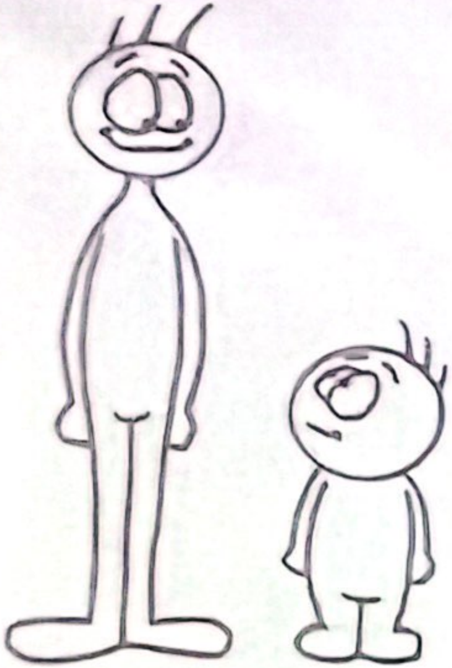
This case involves males with one X and two Y chromosomes in addition to the normal complement of 44 autosomes.

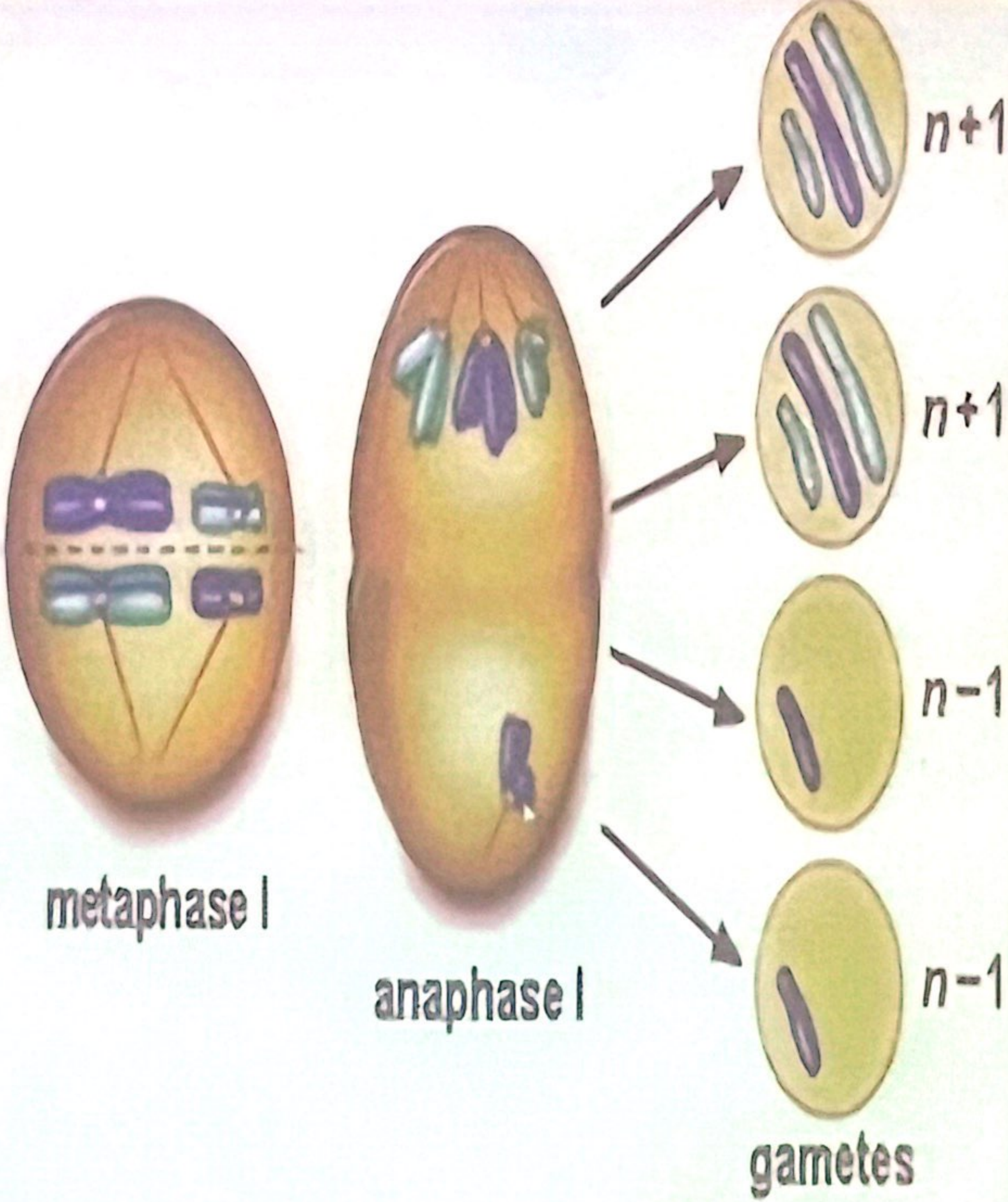
Affected individuals

- taller than average (above 180 cm in length),
- below normal in intelligence,
- usually severely affected with acne.

The frequency condition among criminals was found in some studies to be higher than in normal populations


A1
A2
A3
B4
B5
C6
C7
C8
C9
C10
C11
C12
D13
D14
D15
E16
E17
E18
F19
F20
G21
G22
X?
Y
Y
6




n+1
n+1
n−1
n−1
metaphase I
anaphase I
gametes

4. Down Syndrome (47,21+):

This is the only autosomal trisomy in which a relatively large number of affected individuals live longer than one year after birth.

The syndrome called Mongoloid Idiocy or Mongolism because the affected child has mangol like eye shap

The condition is caused by trisomy of chromosome No. 21.

Children affected are

short , small heads,

idiotic features, broad hands and fingers

mental deficiency.

The incidence about one in every 700 live births.

It originates mostly from nondisjunction in the mother.

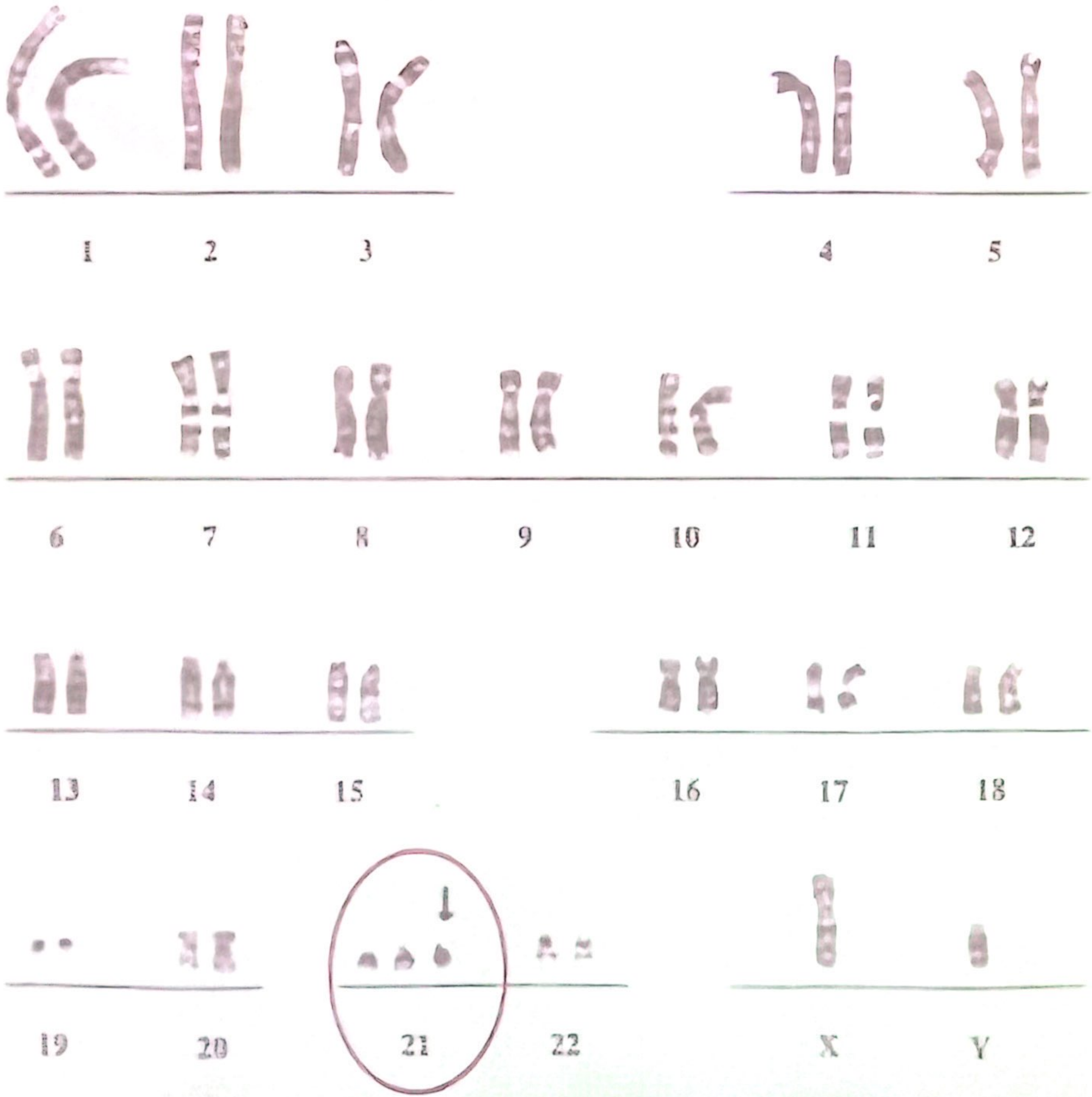


Fig. 4. Trisomy 21 Down syndrome male karyotype (47,XY,+21).

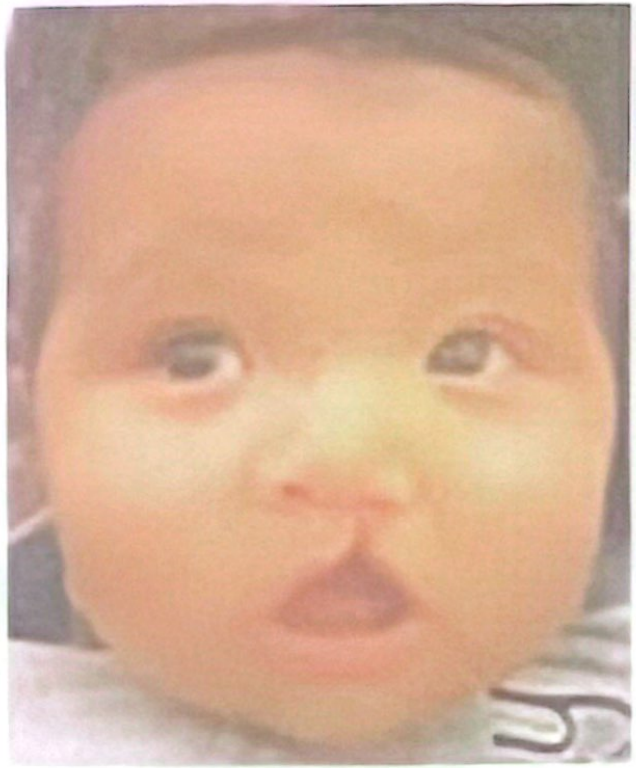
5. Patau Syndrome (47,13+):

This condition is caused by trisomy of chromosome No 13.

Affected infants are born with
severe malformations,
including harelip, cleft palate and polydactyly,
mental retardation.

They usually live for less than six months.

It is a rare condition, with an incidence of about one in every 20,000 births.

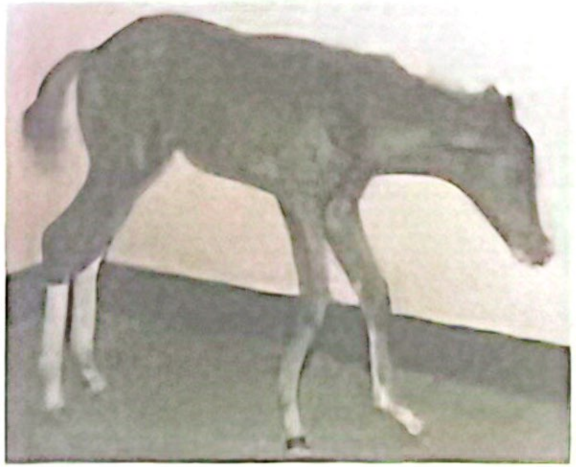


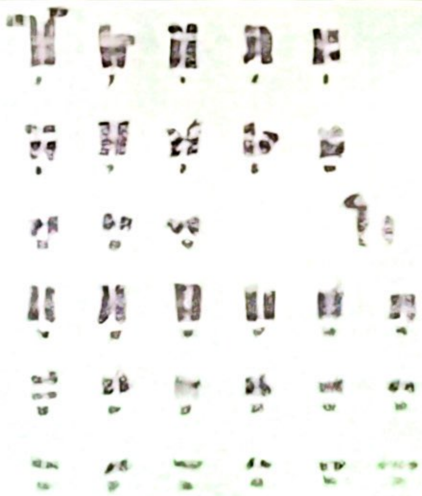
Patau syndrome

Trisomy 31 (65, 31+,xy)

Autosomal trisomy: 65,XY, 31+

Symptoms

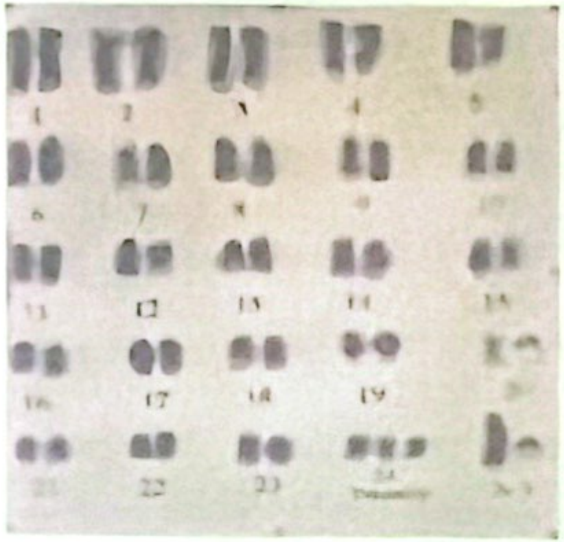




- Limb deformities,
- Severe brachygnathia inferior (parrot mouth)
- Failure to grow
- Genital abnormalities

Trisomy in buffalo bull (51, 24+,xy)

Trisomy for **chromosome 24** as possible cause of testicular hypoplasia in buffalo bulls



Symptoms

- The affected bull had azoospermic semen from puberty

Darwish et al., 2017

In general trisomics of large autosomes and monosomics for any autosome are observed in living humans and animals.

The reduced viability of monosomy or trisomy is an indication that other aneuploid conditions may rise.

This has been confirmed by karyotypic analysis of spontaneously aborted fetuses.

About 30% of all spontaneous abortuses demonstrate some form of chromosomal aberration,

65% of these are aneuploids.

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65% of these are aneuploids.

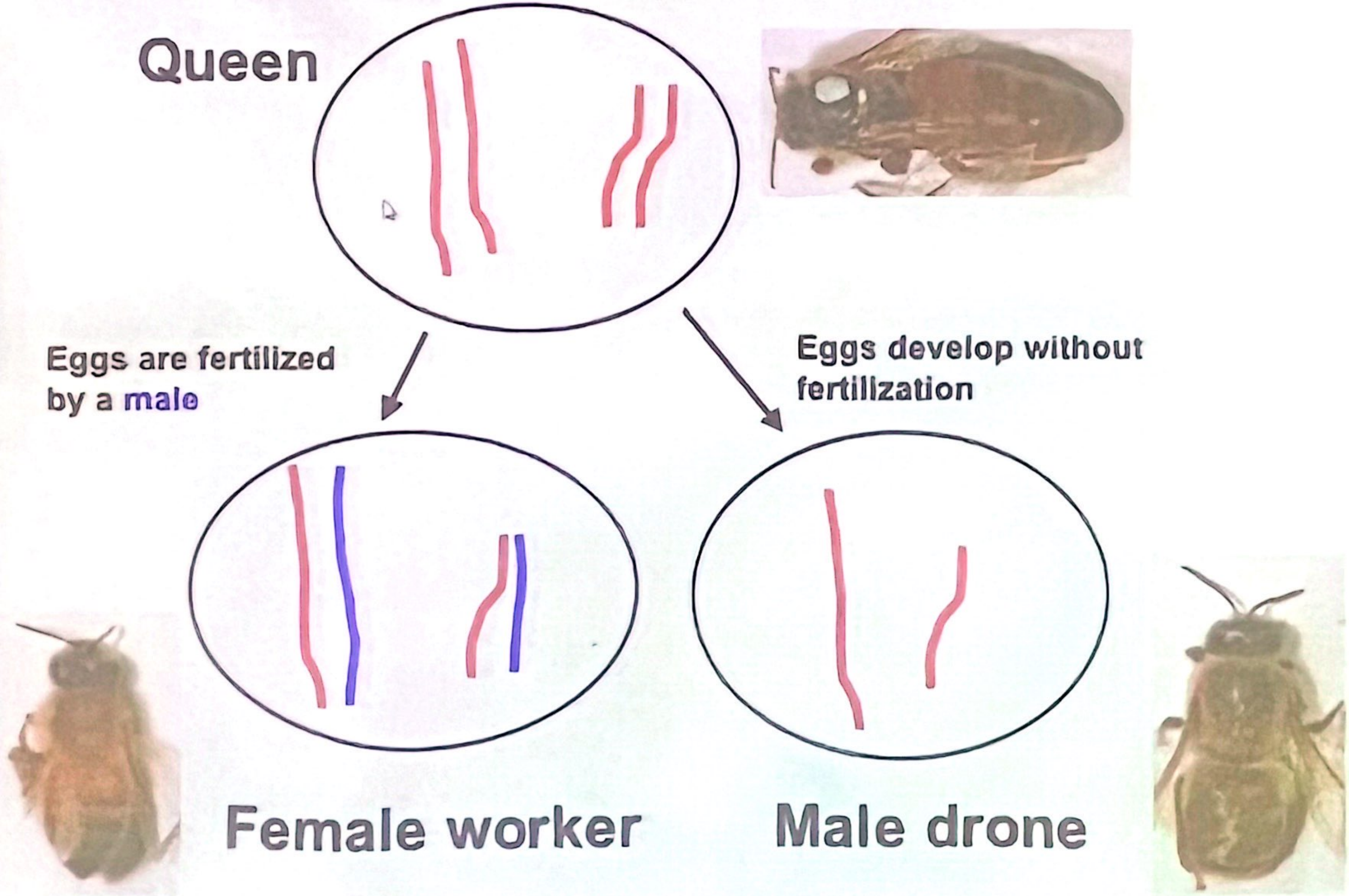
Studies have revealed cases of trisomy in all human chromosomes except No 5, 12 and 17.

III- Monoploidy:

This is synonymous with haploidy (N).
This condition is normally present in male bees and wasps.

It results from the development of the ovum without fertilization, which is called parthenogenesis.

Idealized diagram of chromosomes in cells of bees



V. Polyploidy:

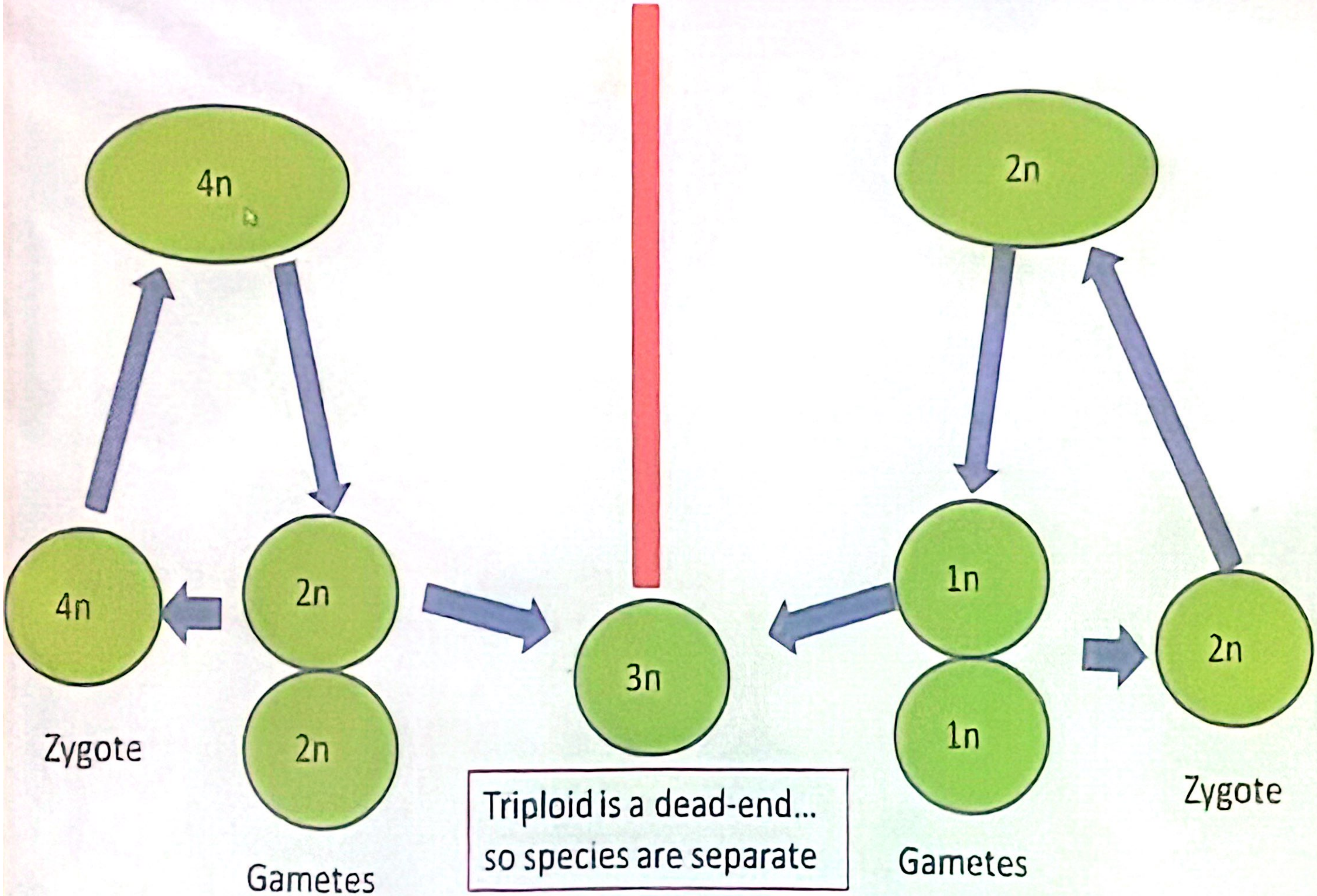
The term polyploidy cases more than two haploid chromosome sets.

triploid has three sets (3n);
tetraploid has four sets (4n);
pentaploid has five sets (5n);
hexaploid has six sets (6n),

Polyploidy common in plants than animals.
It is well known in fish, amphibians and lizards.

Polyploids with odd numbers of chromosome sets (e.g. triploids 3N and pentaploids 5N) fail to transmit the condition from one generation to the next.

Genetically balanced gametes cannot be produced, leading to sterility of odd polyploids.


4n
2n
4n
2n
2n
Zygote
Gametes
3n
Triploid is a dead-end...
so species are separate
1n
1n
2n
Gametes
Zygote

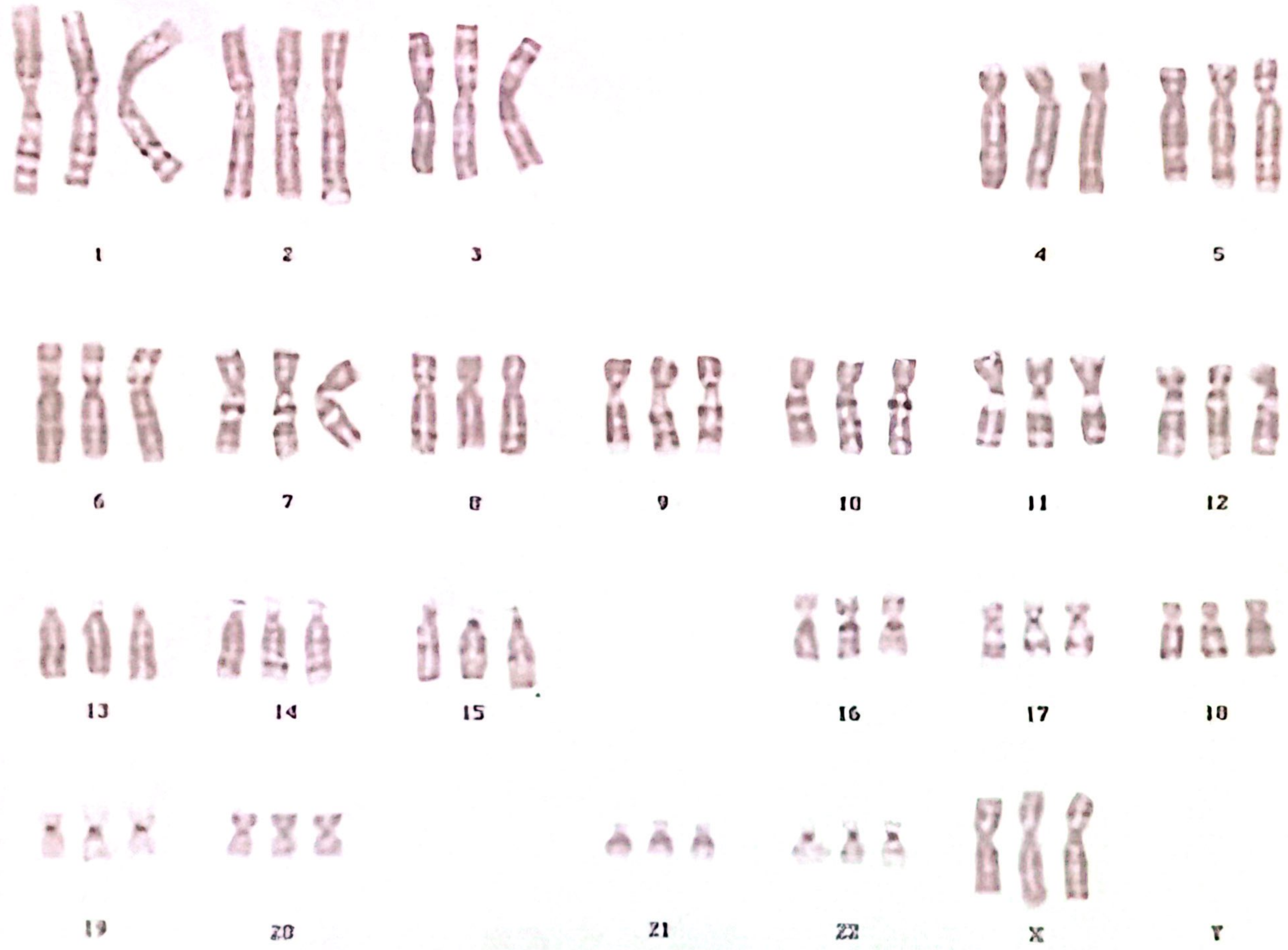
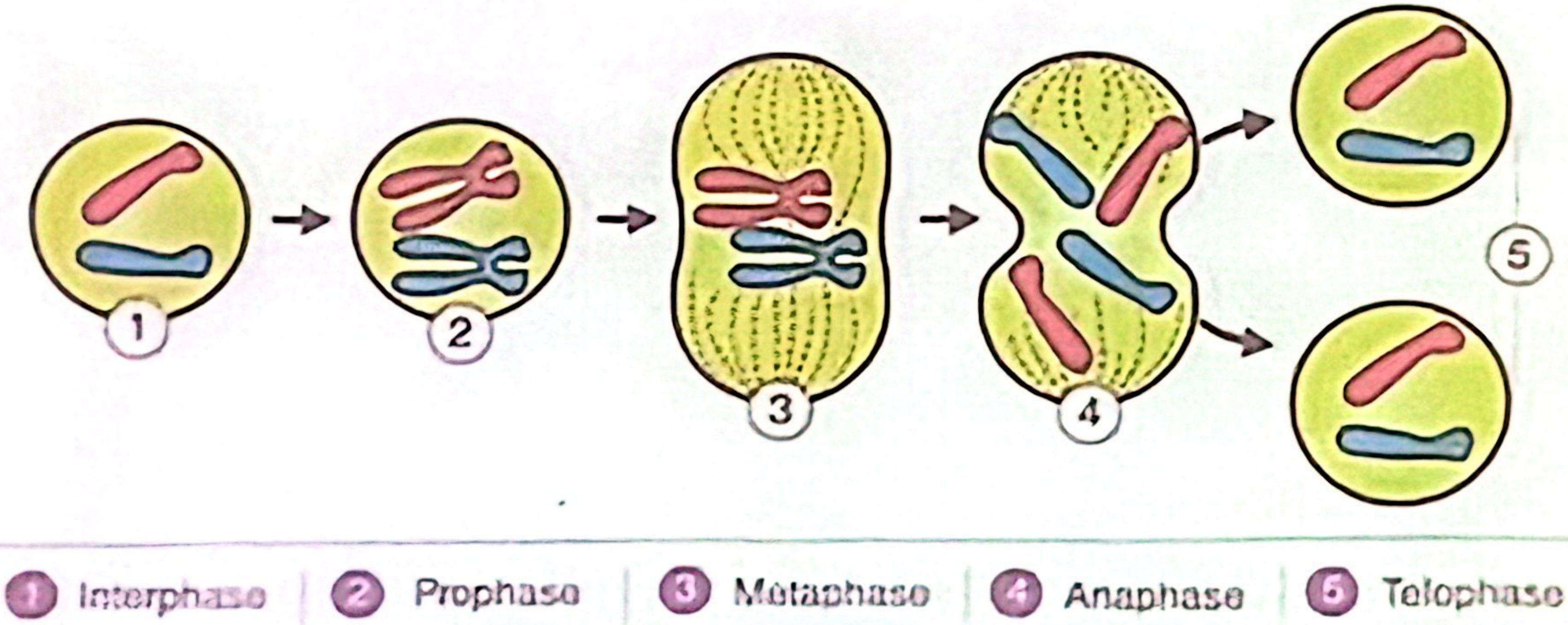


Fig. 9. Karyotype of a triploid fetus (69,XXX).

V.1- Endopolyploidy:

A process by which the chromosome replicate without nuclear division of the nucleus



V.1- Endopolyploidy:

Certain cells in a diploid organism have been observed to be polyloid.

The process which leads to endopolyploidy is called *endomitosis.*

Vertebrate liver cell nuclei, including human ones are often endopolyploidy, with 4n, 8n, or 16n chromosome sets.

Liver cell are very active cells in a diploid organism,

it may be necessary to replicate the entire genome by endopolyploidy allowed increased rate of expression of various genes.

Structural Chromosomal Changes

Changes in the internal structure or arrangement of genetic material within or between individual chromosomes

This change occur as a result of:

↓

Chromosomal breakages and reunion of the broken ends in new way

- If both chromatids affected (whole chromosome arm) $\xrightarrow{\text{called}}$ Chromosomal aberration

- If only one chromatid affected $\xrightarrow{\text{called}}$ Chromatid aberration

Structural Chromosomal Changes

Balanced Structural Rearrangements

1- Inversion
2- Shifts
3- Reciprocal translocation
4- Centric fusion
5-Tandam fusion
6- Insertion
7- Transposition or non reciprocal translocation

Unbalanced Structural Rearrangements

1- Deletion
2- Duplication
3- Isochromosomes
4- Ring chromosome

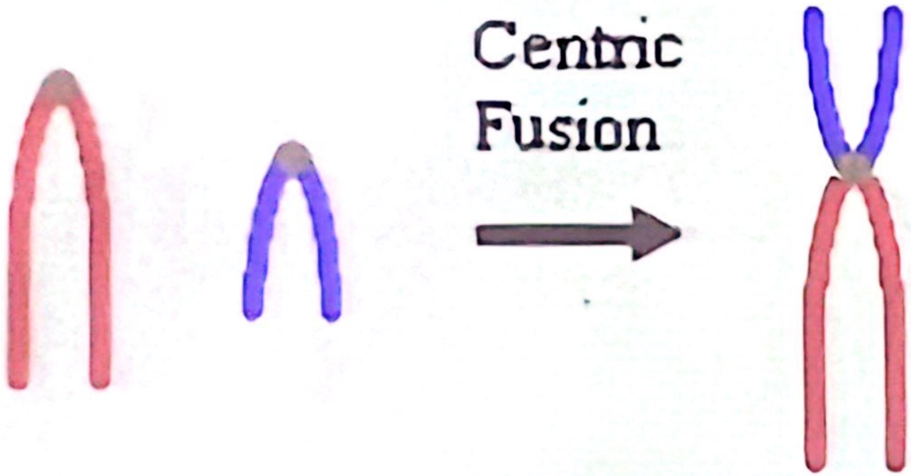
Centric fusion (Robertsonian translocation)

- Type of translocation between two non homologues acrocentric chromosomes

- Centromere of two acrocentric chromosomes fused to produced

one metacentric or

submetacentric chromosome

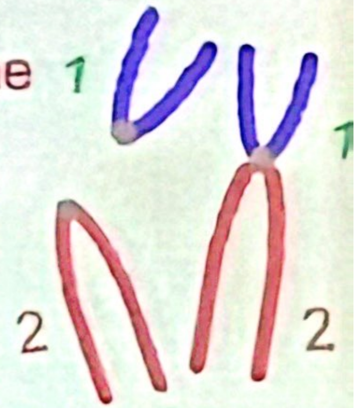


❖ Fused chromosome contain all the genetic material previously present in the two separate chromosome

Centric fusion (Robertsonian translocation)

- The fused chromosomes contains all the genetic material present in two separate chromosome.......Individal have a complete genome and have normal phenotype

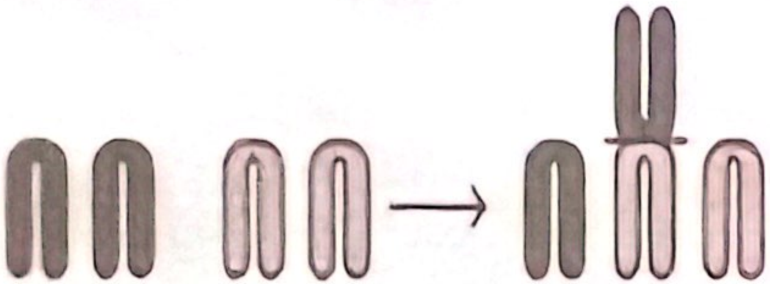
- In heterozygous centric fusion , (replacement of two chromosome by one)...........The total number of chromosome is one less than the normal number.



In human 46 reduced to 45 ➢

- In homozygous centric fusion have two chromosome less than the normal number of the chromosome
- Also have normal phenotype ....have a complete genome.

Centric fusion (Robertsonian translocation)

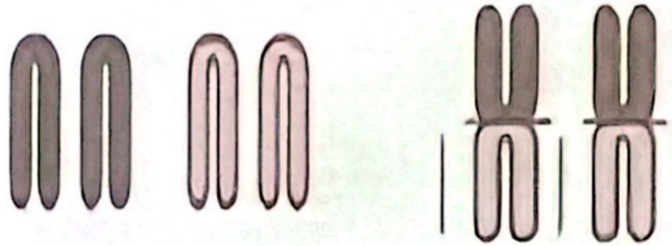


Heterozygous for centric fusion

Replacement of two chromosome by one

one chromosome less than the normal number

46 ⟶ 45



Homozygous for centric fusion

Replacement of four chromosome by two

two chromosome less than the normal number

46 ⟶ 44

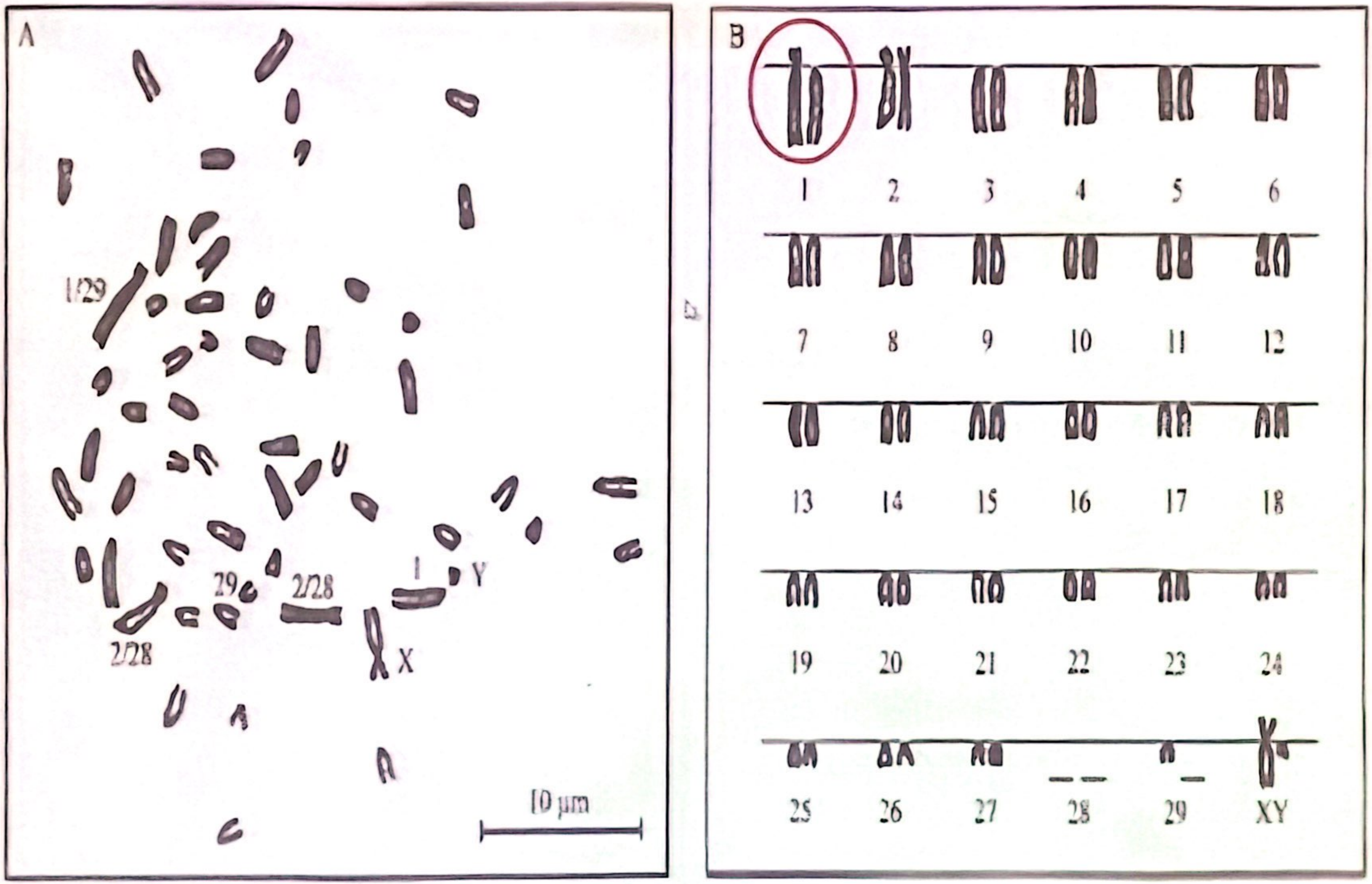
Unlike monosomics, individuals heterozygous for a centric fusion have a complete genome and have normal phenotypes.

Effect of centric fusion on fertility

- **In sheep** → Ram heterozygous for Robertsonine translocation produce 5% unbalanced Spermatocyte → not survive to become unbalance gametes

- **In cattle** → unbalanced gametes able to fertilized or produce trisomic and monosomic embryo → subsequently survive

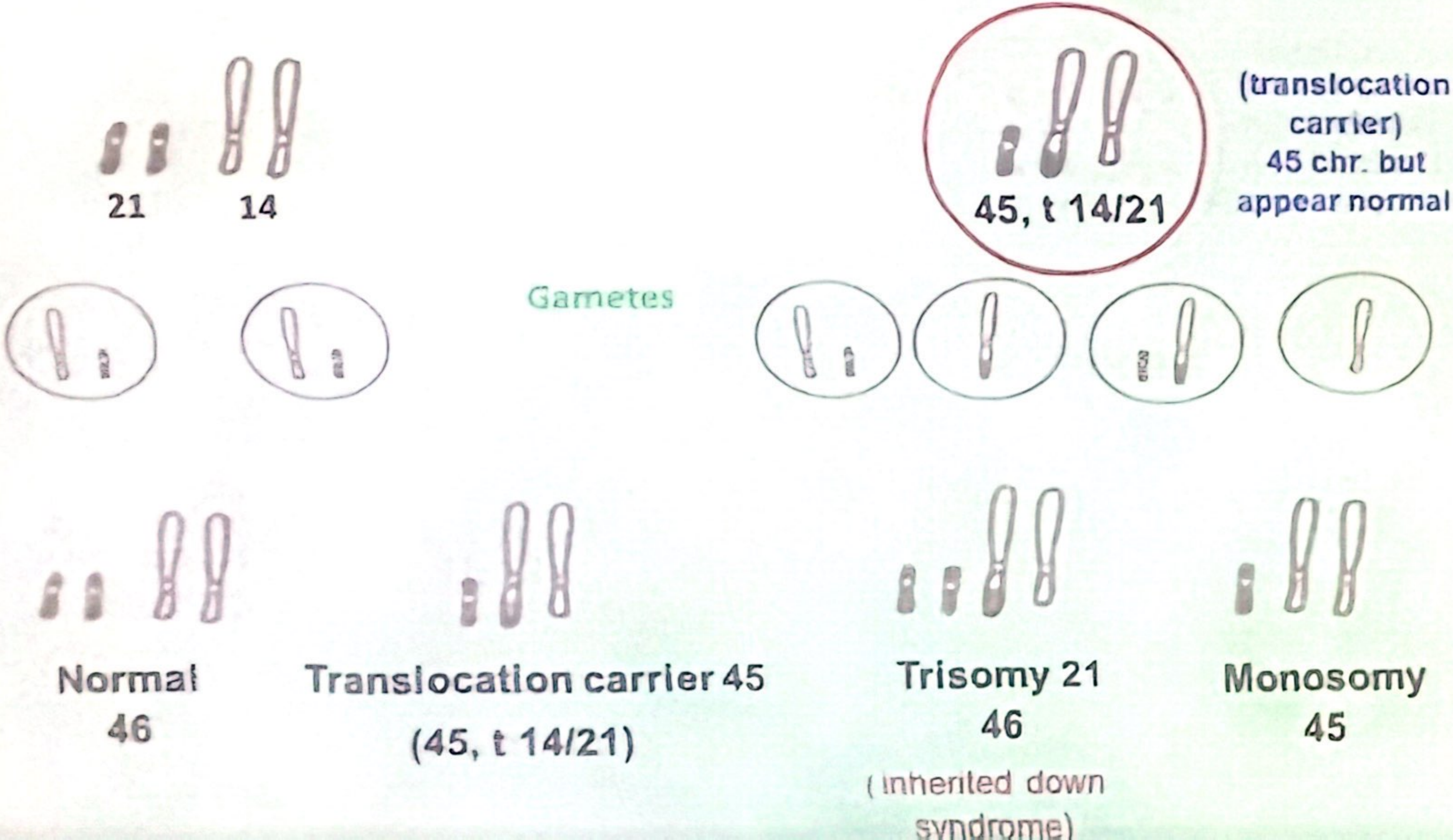
In cattle, most commen translocation, 1/29 (59,t 1/29) → normal individule with lower fertility (due to lethal condition in zygotes or develping embryo)



Centric fusion in cattle

Ex. Familial or Inherited Down Syndrome

Cause: Robertsonian translocation between chromosomes 14 and 21 45,t (14/21) in a heterozygous condition in one of **parent** of affected child



Diffrences between Familial down syndrom and Down syndrom

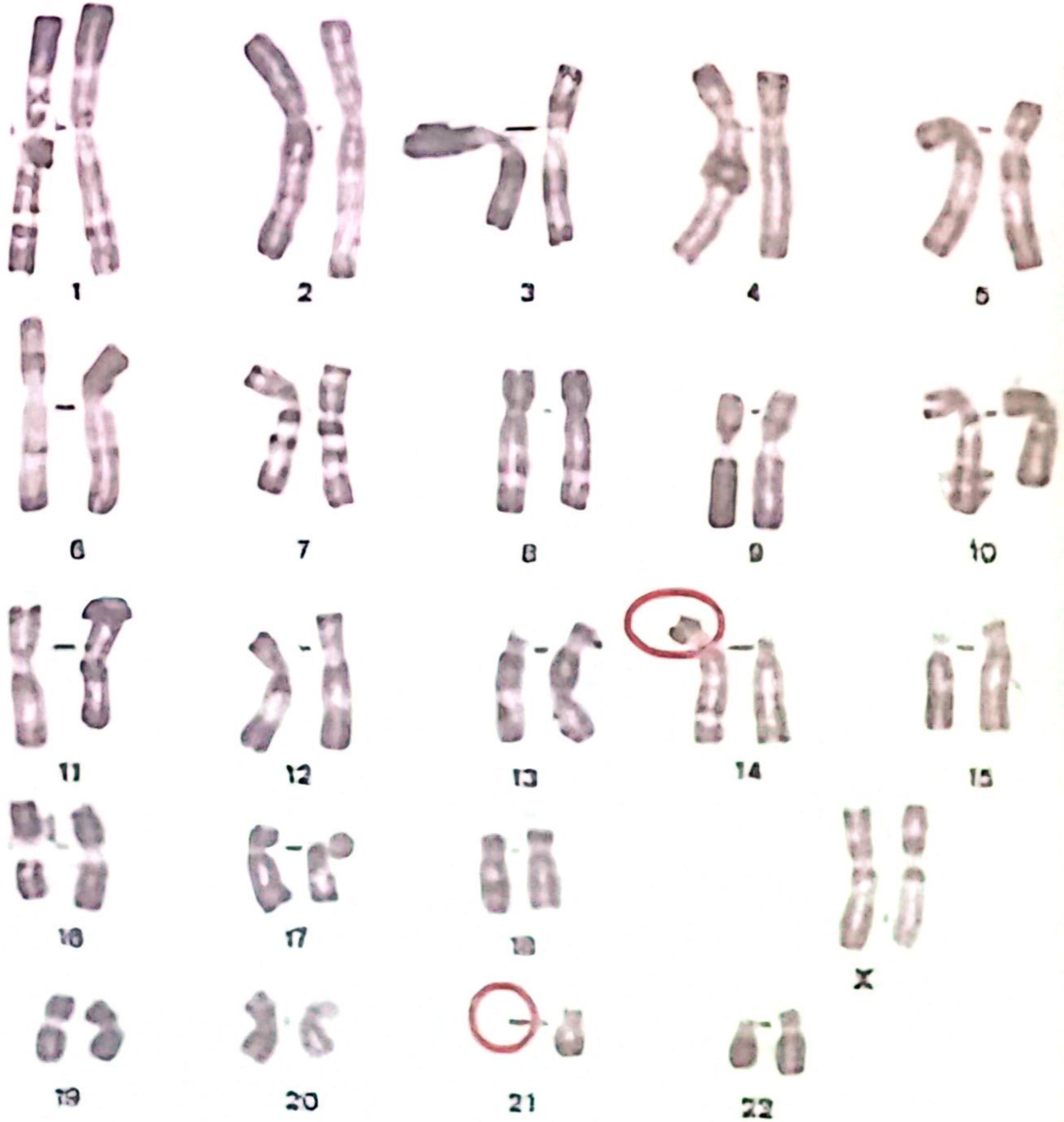
Familial down syndrom	Down syndrom
- Inherited	- Non Inherited
- Due to centric fusion between chromosome 14 and 21	- Due to non disjenction in mother gamate
- 46,t 14/21	- 47, 21 +
- Balanced Stractural chromosoal aberration	- Numerical chromosoal aberration
- 5 % of all the caces	- 95 % of all the caces

45,XX,rob(14;21)(q10;q10)

Loss of the short arm(s) of chromosomes 14 and 21

Normal phenotype

carrier 45, t 14/21



Down Syndrome 47, 21 +

Unbalanced Structural Rearrangements

1- Deletion

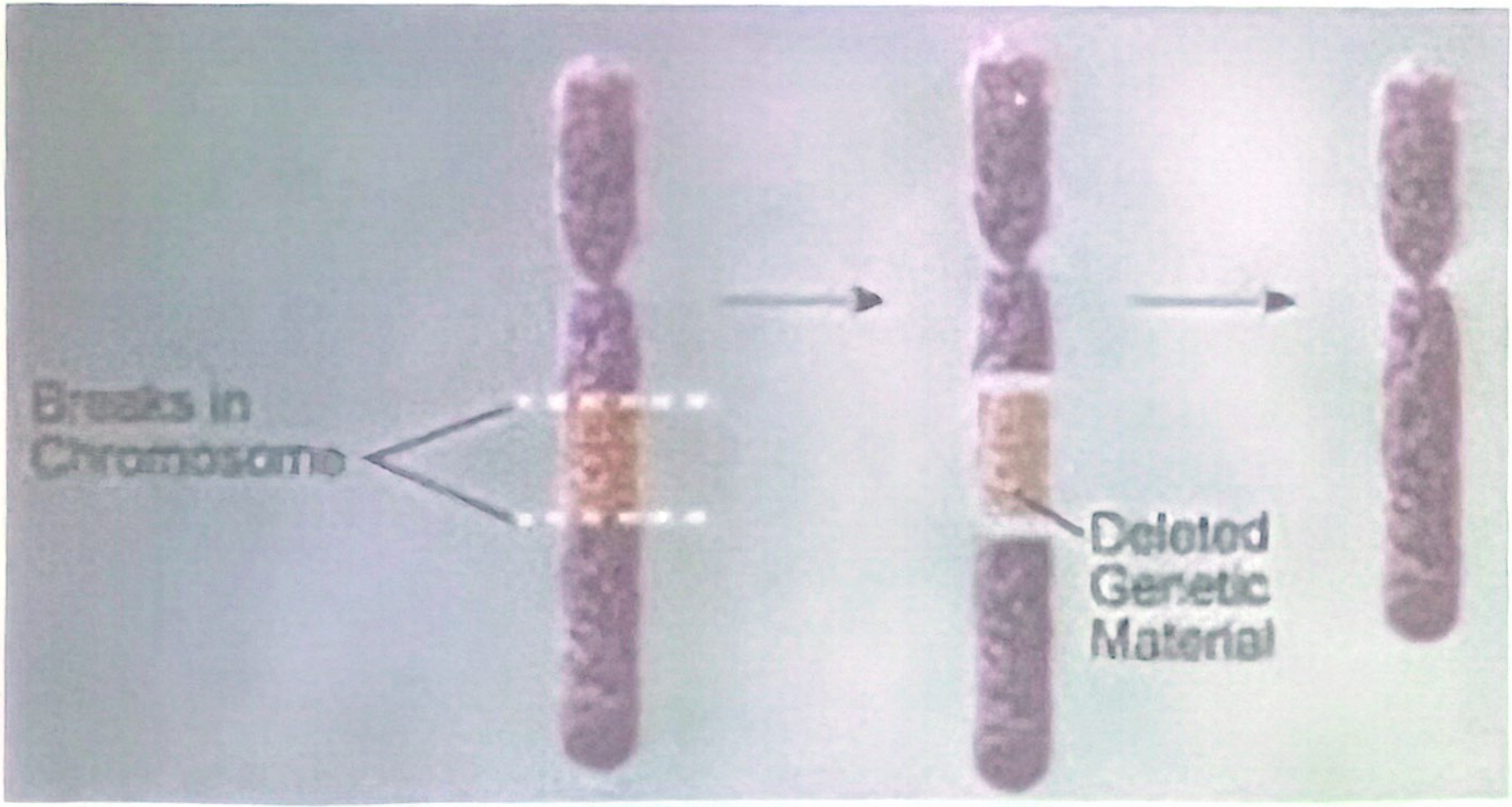
2- Duplication

3- Isochromosomes

4- Ring chromosome

1- Deletion

- Occur when a chromosome loses a segment of its original material

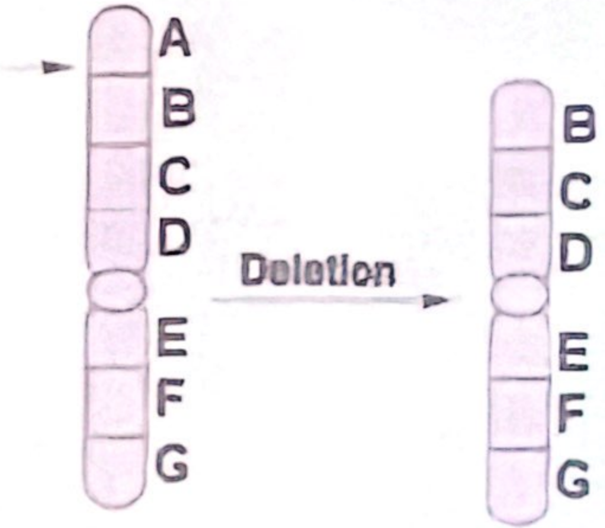


1- Deletion

Types of deletions

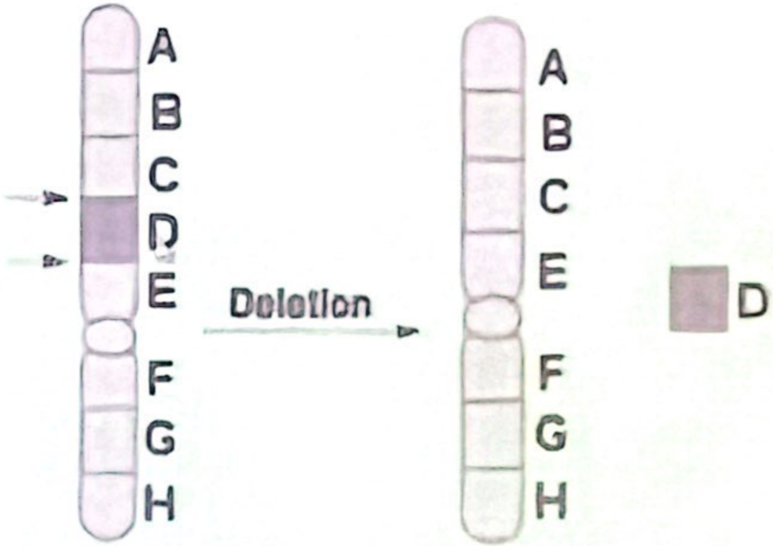
Terminal deletion

if the lost segment is at the end of the chromosome (include telomere)



Interstitial deletion

if two breaks occur and unite leaving out the segment



1- Deletion

- Deletions result in fragments without a centromere (acentric) which will be lost at the next cell division.

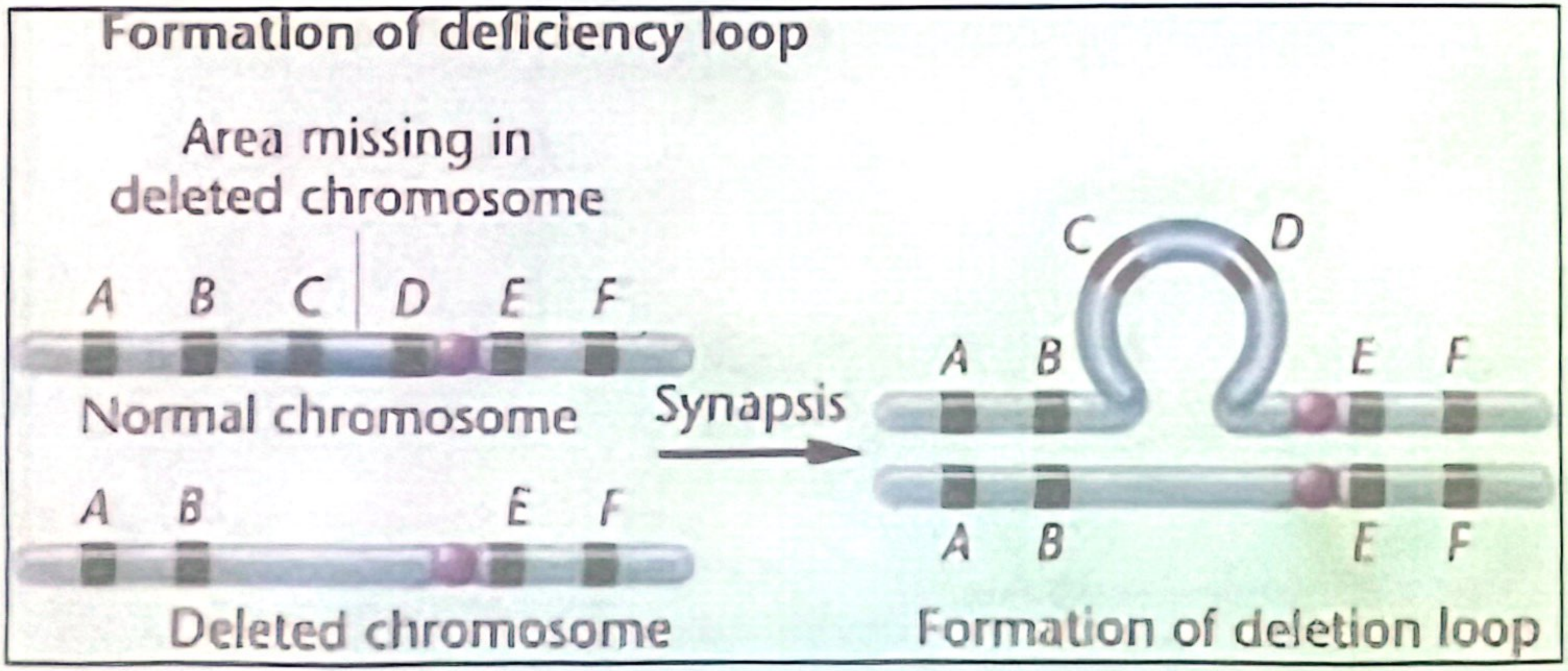
- Deletions may lead to an alteration in the phenotype depending on whether an autosome or a sex chromosome is affected

Deletion during meiotic division

Pairing or synapse of homologues chromosome in case of heterozygous deletion

↓

Compensation loop (deletion loop) is formed in intact chromosome



1- Deletion

Ex. Cri-du-Chat Syndrome (46,5p-)

Causes → due to a heterozygous deletion of about half the smaller arm of chromosome **5** (5P-)

Symptoms

- Many anatomical malformations
- Abnormal development of the glottis and larynx
- Infants give a characteristic cry similar to that of the meowing of a cat
- Severely mentally retarded
- Cri-du-Chat syndrome is about 1 in 50,000 live births

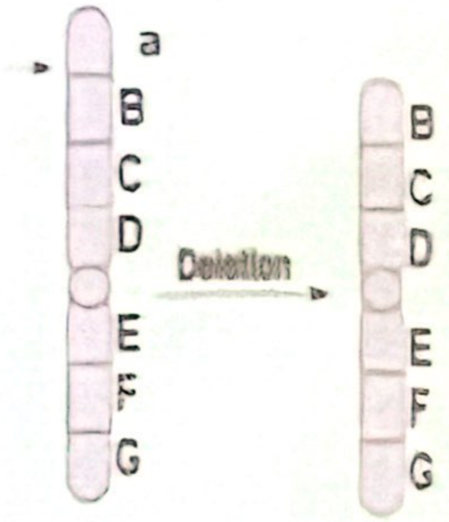
Effect of Deletion on Phenotype

- **Heterozygous deletion** in the region containing a certain dominant gene while the normal homologue contains its recessive allele will lead to a **hemizygous condition**

↓

Recessive gene act as a dominant gene and this called

pseudo-dominance

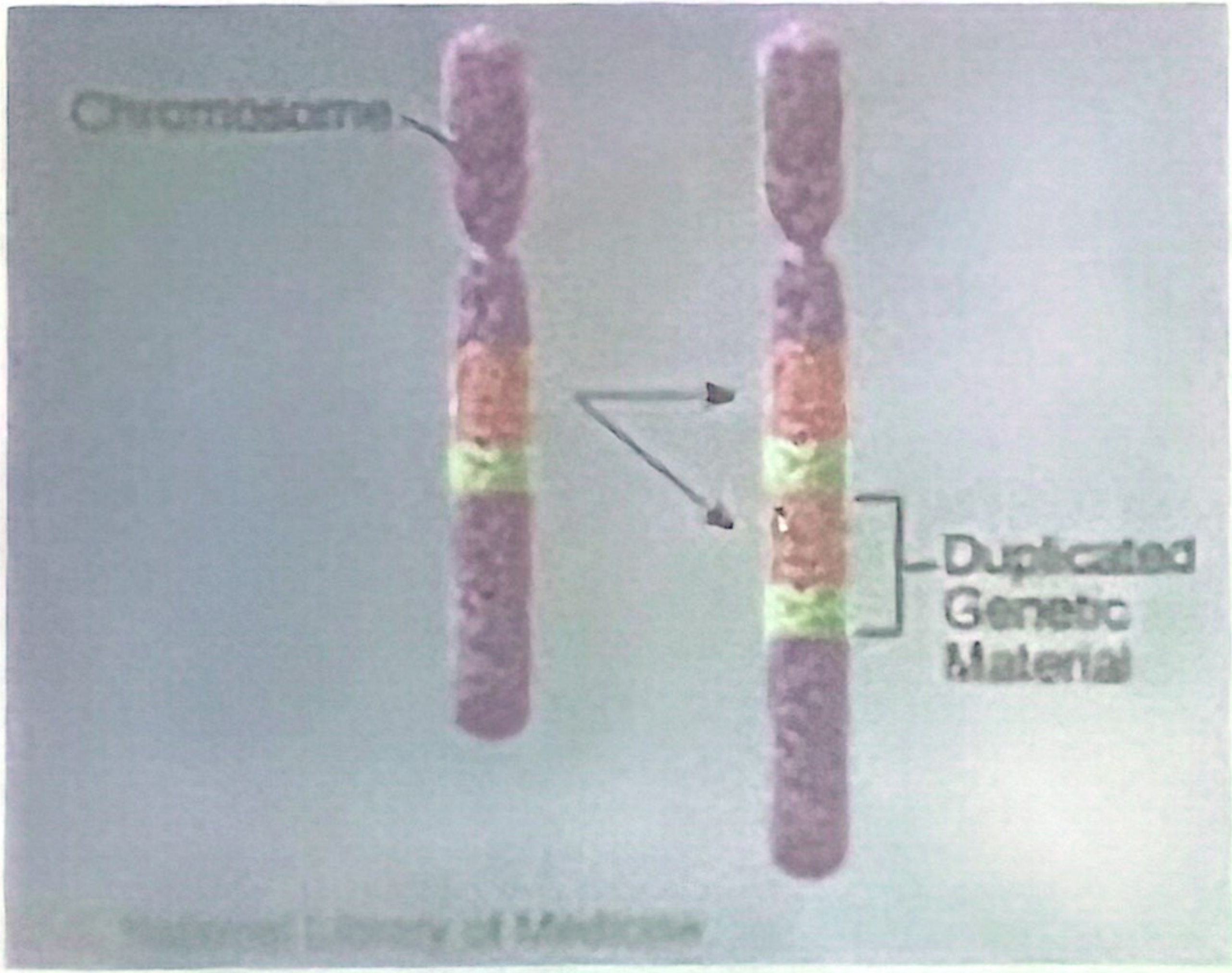


- A homozygous deletion is supposed to be lethal

- Deletion in the X chromosome of a male (XY) is lethal

2- Duplication

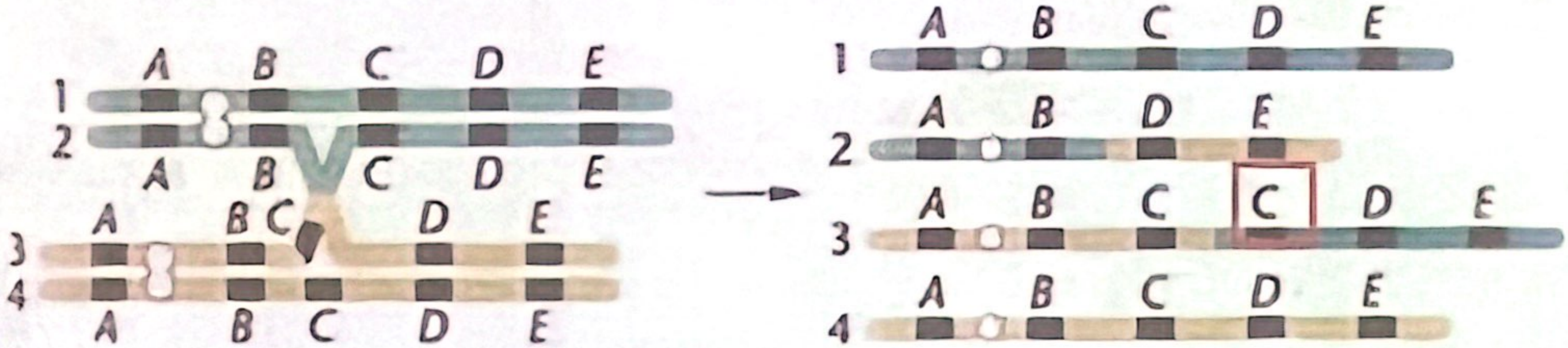
Occur when any part of the genetic material, whether a single locus or a large piece of a chromosome, is duplicated



2- Duplication

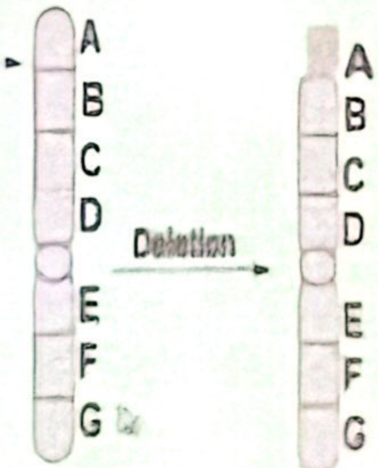
Causes

1- Unequal crossing over between synapsed chromosomes during meiosis



2- Replication error before meiosis

3- Through insertion of a deleted fragment into homologous chromosome



2- Duplication

Types of Duplication

Tandem

Revers tandem

Normal Sequence

centromer

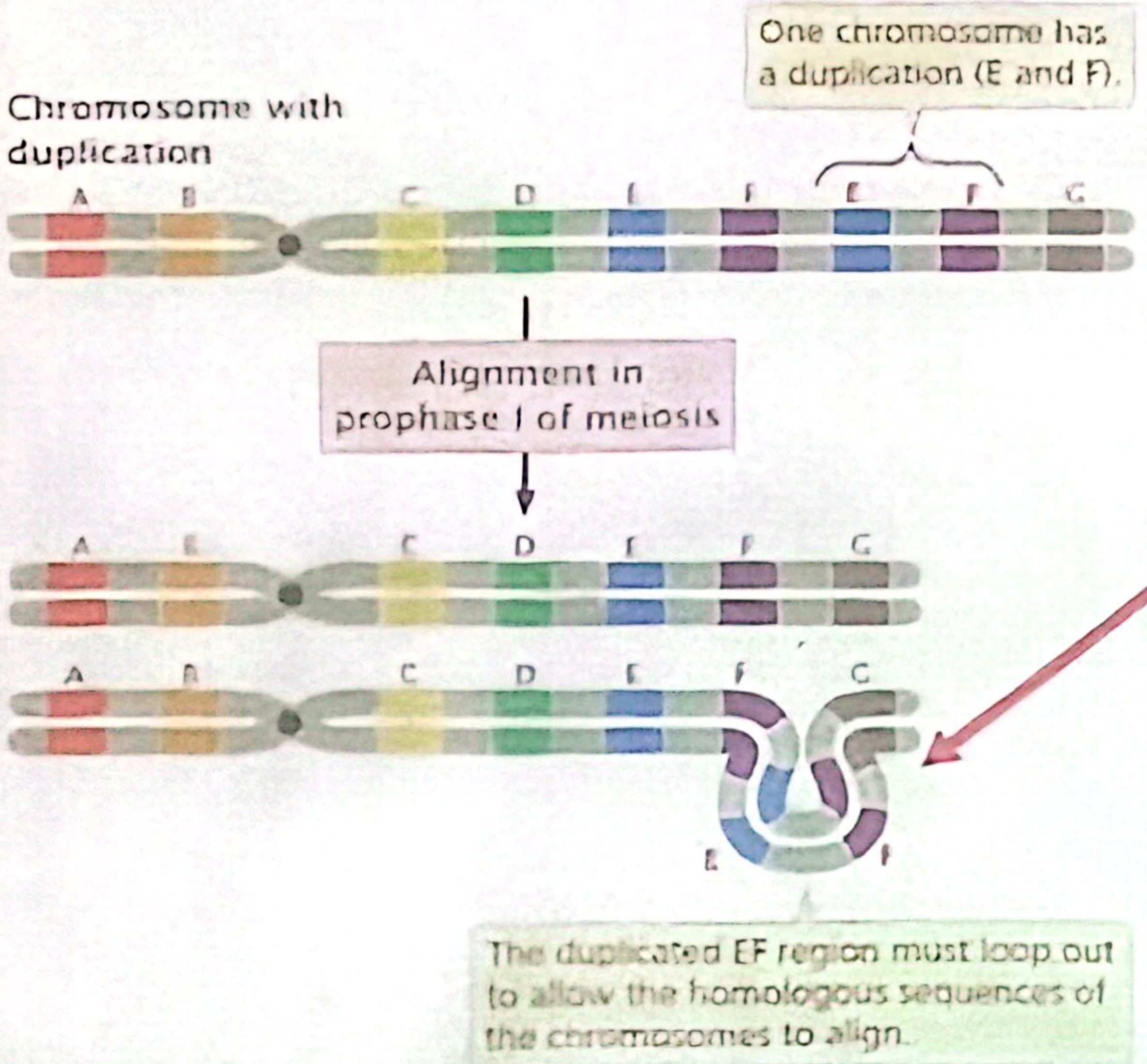
a b c d e f g h i j

a b c d e d e f g h i j

Tandem

Duplication during meiotic division

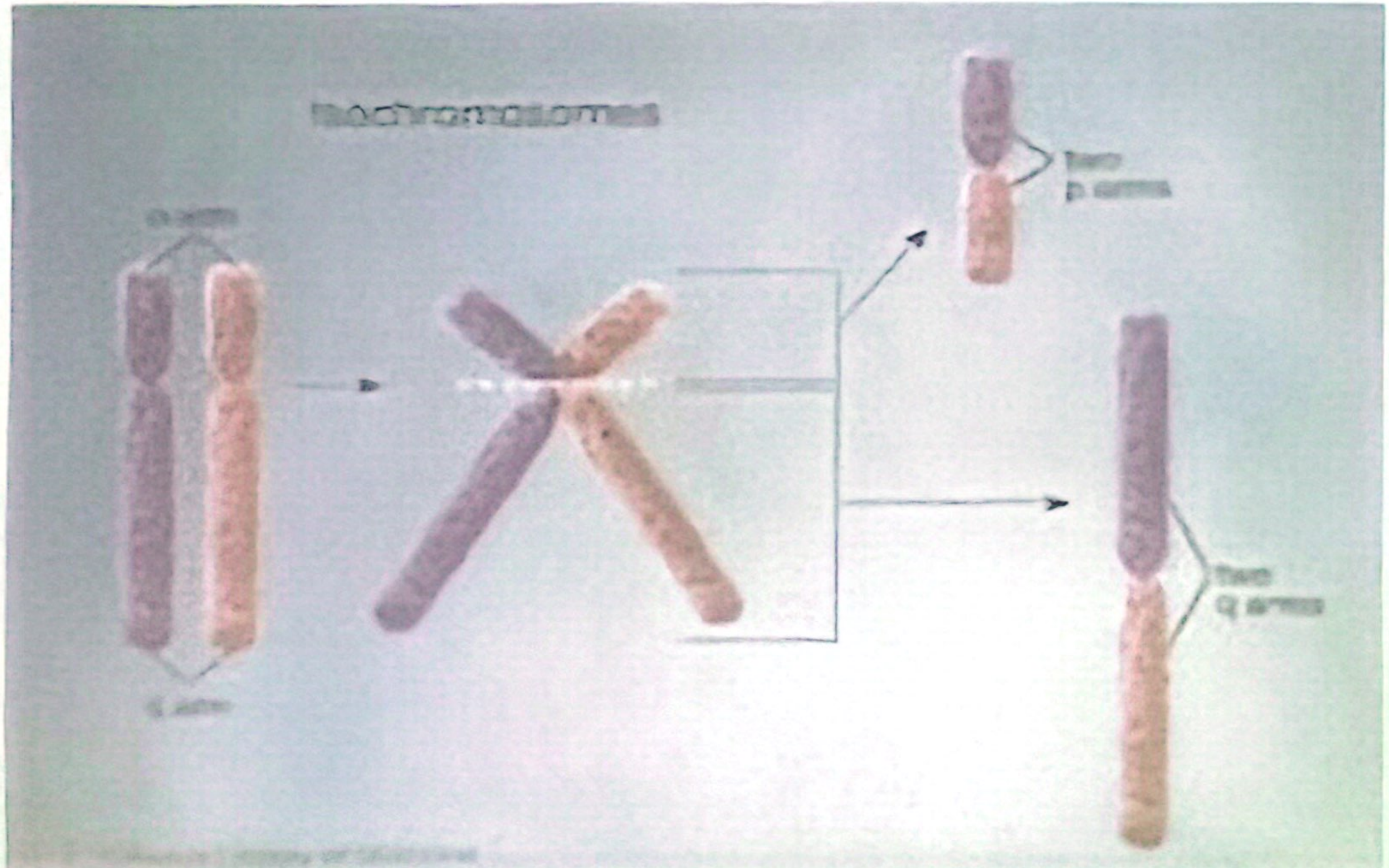
Pairing or synapse of homologues chromosome in case of heterozygous duplication



Compensation loop (duplication loop) in affected chromosome

3- Isochromosome

- Are formed due to an error involving the centromere at anaphase
- Normally, the centromere splits along the long axis of the chromosome and homologou parts of sister chromatids move to opposite poles.



4- Ring chromosome

Formed if a terminal deletion occurs in both arms of a chromosome and the broken ends reunite